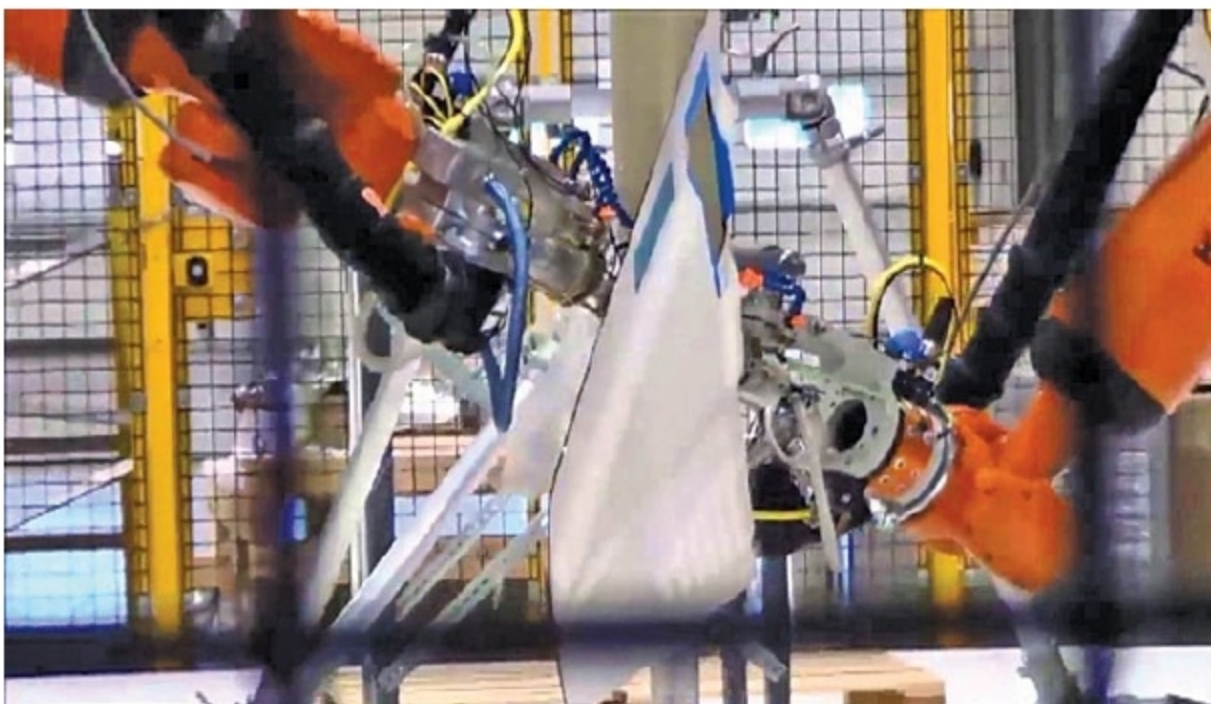




GE's HydraStar system (Credit: Youtube)

Earlier, a skilled workforce was needed for timely inspections and analysis of results. With growing industry sizes, it is practically impossible to track all assets manually, and even the inspection teams cannot cover all areas to be able to identify the source of malfunctioning of a machine. Inspection processes we had in past were unreliable, time-consuming and unsafe for humans. To address these challenges at the industrial scale, smart inspection solutions are taking over.

Automating asset inspection

The basic idea here is to collect the most relevant data of machine health and transfer it to a cloud platform in real time, where it can be processed to provide owners with understandable results. Users can access these data through smartphones, tablets or laptops in real time. Insights and reports can be referred to and understood in terms of which part of the machine needs to be repaired.

Technologies like the Internet of Things (IoT) have made it possible for machines to be able to talk to us and to each other. There are sensors embedded within machines, linked with applications, which can provide real-time results. There are different kinds of solutions available today, using various technologies, that can be used for the purpose of industrial asset monitoring. We discuss a few below.

Visual inspection by drones. One can now have automated, unmanned

drones with scheduled flights in line with required environmental and weather conditions. These solutions can automatically recognise potential inspection issues and create a report of issues along with their location. This report is delivered automatically to subject matter experts for review, who can take action for repairs.

One industry that is extensively using drone technology is insurance. Large-scale operations insure technical assets. During damage claims, insurance companies require verification of the claims to avoid frauds. Drones technology has enabled insurance companies to remotely monitor the site of the claims. It has also reduced the risk to employees who perform verification on site.

As per a study by Deloitte, drones increase inspection efficiency by 85 per cent and, in the process, duration of claim settlement has reduced from days to hours. Drones technology can also help counter fraudulent funds.

Other industries that can benefit from drone technology include oil and gas, electricity, railways, solar and wind, and so on. Big players like Deloitte are providing drone technology visual inspection solutions. Many Indian startups like Detect Technology are also delving into this area.

Ultrasonic inspection by robots. Robots have

ultrasonic sensors that act as their eyes and enable them to detect obstacles and walls. These reduce manual work and can inspect inaccessible or hazardous areas without risking human lives. Robots allow cheap, efficient and safe inspections in critical industries such as nuclear and petrochemical, and in a variety of other use cases like inspecting pipelines in small industries, monitoring machinery in large industries like aerospace and so on.

General Electronics (GE) has come up with an ultrasonic inspection technology, termed as HydraStar dual robotic ultrasonic inspection system. A major manufacturer in south-eastern Georgia, USA, produces structural components of aircrafts, including fuselages, wings, empennages, nacelles and helicopter cabins, for commercial, military and business jet aircraft industries. For inspection, the company used non-destructive testing and flatbed scanners for nearly 15 to 20 years. GE's HydraStar system reduced part inspection time by 33 to 43 per cent, depending on the part. With GE solutions, the manufacturer was able to achieve faster inspection with fewer machines. Inspections were much flexible and cost-effective, too.

Two major players providing such solutions are GE and Toshiba Energy Systems (specific to generators). In addition to these, there are multiple other ultrasonic robot inspection solutions available in the market.

Industrial IoT (IIoT) for inspection. Using traditional methods, people can be notified only when there is an abnormality, which is not the ideal way to combat system



Electricity line inspection by drones (Credit: ABJ Drones)